

# FraudShield BI

## Quantifying the Business Impact of False Positives

**91.14%**

Precision

**78.18%**

Recall

**84.17%**

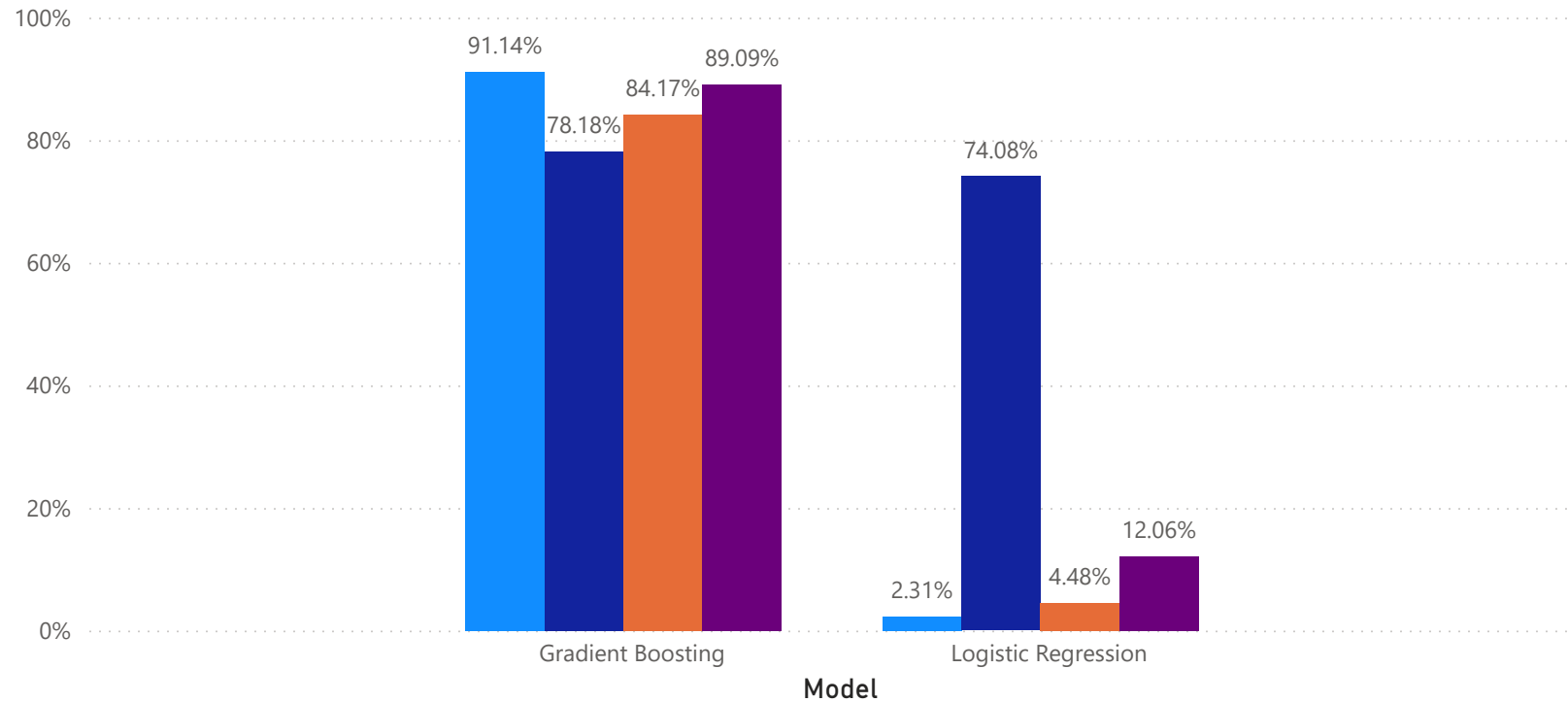
F1-Score

**163**

False-Positives

### Model Performance Comparison

● Precision ● Recall ● F1-Score ● Pr-Auc





Executive Overview

Threshold Decision Lab

Cost & Recommendation

False-Positive Deep Dive

Model

Threshold

Gradient Boosting

Logistic Regression

0.10

0.20

0.30

0.40

0.50

0.60

0.70

0.80

0.90

91.14%

Precision

78.18%

Recall

84.17%

F1-Score

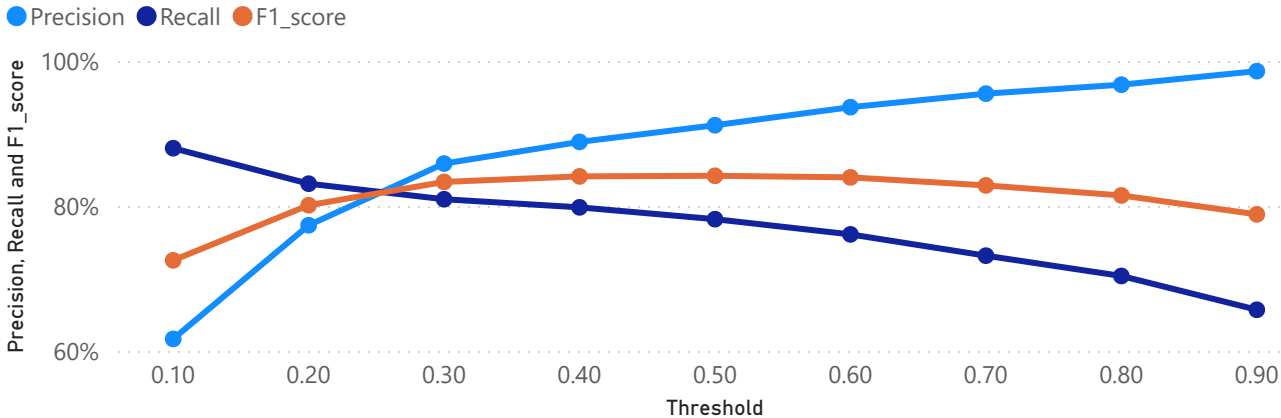
163

False Positives

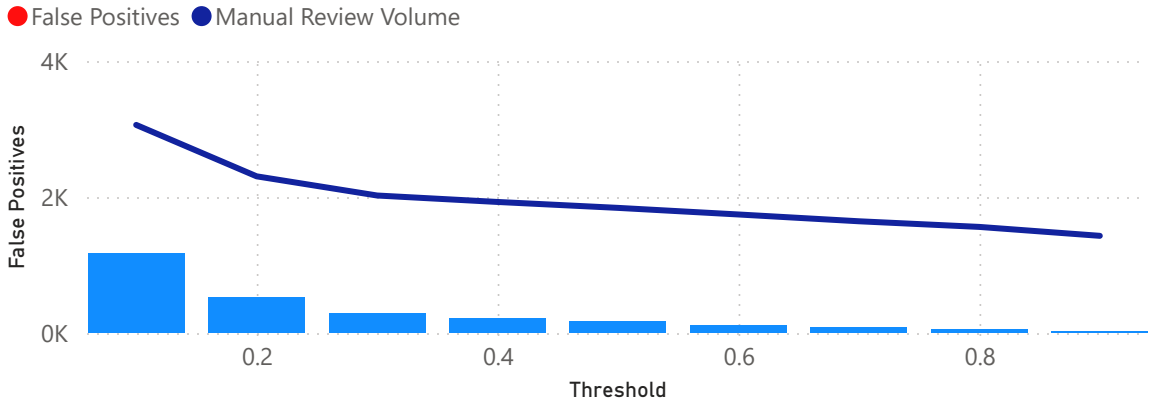
1,840

Manual Review Volume

Threshold Trade-Off: Precision, Recall and F1-Score



Business Impact by Threshold

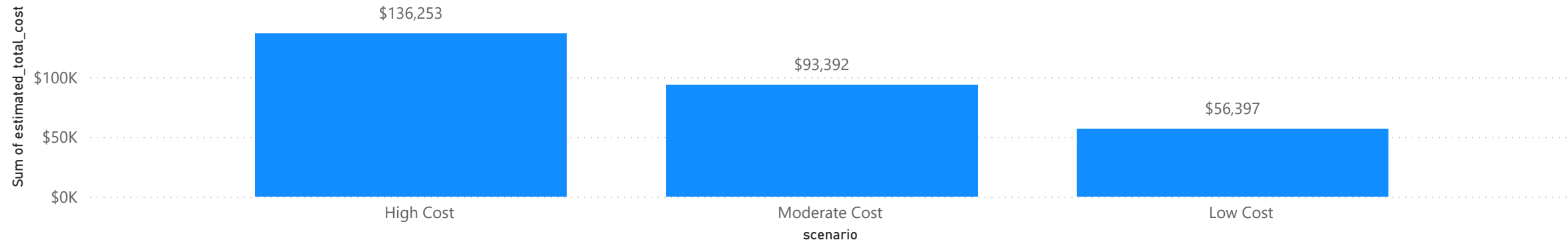




Cost Scenario

| High Cost                                                 | Low Cost                                         | Moderate Cost                                       |
|-----------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| <div>Gradient Boosting</div> <div>Recommended Model</div> | <div>0.10</div> <div>Recommended Threshold</div> | <div>\$93,392</div> <div>Estimated Total Cost</div> |
| <div>1,173</div> <div>False Positives</div>               | <div>258</div> <div>False Negatives</div>        | <div>\$50,532</div> <div>Missed Fraud Amount</div>  |

Estimated Total Cost by Scenario



Recommended Decision

Under the Moderate Cost scenario, Gradient Boosting at a threshold of 0.10 is the recommended decision. This setting produces an estimated total cost of \$93,392, with 1,173 false positives, 258 false negatives, and \$50,532 in missed fraud. The result reflects the cost assumptions used in this analysis and provides the lowest estimated cost among the evaluated model and threshold combinations

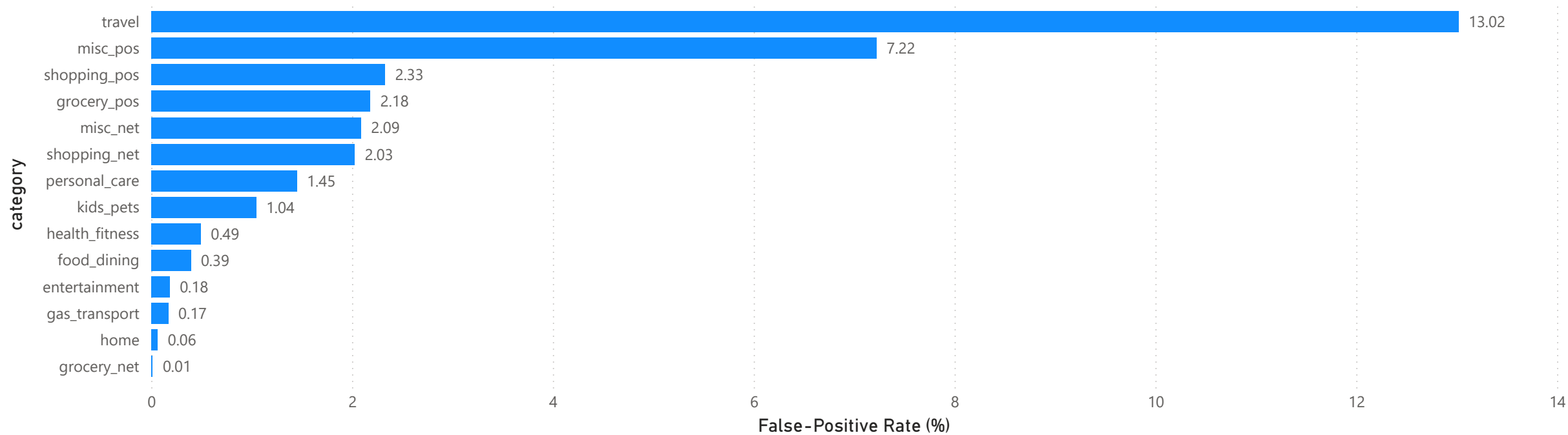
Executive Overview

Threshold Decision Lab

Cost & Recommendation

False-Positive Deep Dive

False-Positive Rate (%) by category



Fraud Rate (%) by Transaction Hour

